DRONE VOLT®

AERIALTRONICS

User Manual

Crowd Counting

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1. Introduction

The Crowd counting model uses a deep learning neural network to estimate the number of people appearing in an image with high accuracy and high processing speed.

The neural network contains many convolution layers to understand images. It takes an image (or a video) as input and it outputs a density map of the human head. "*Figure 1*" illustrates an example of the output of a processed input image. The brighter color on the output map indicates the higher density of the human head. Therefore, the estimated number of people appearing in the crowd is obtained based on the generated density map. For example, the estimated counting value is "512" for the below image.

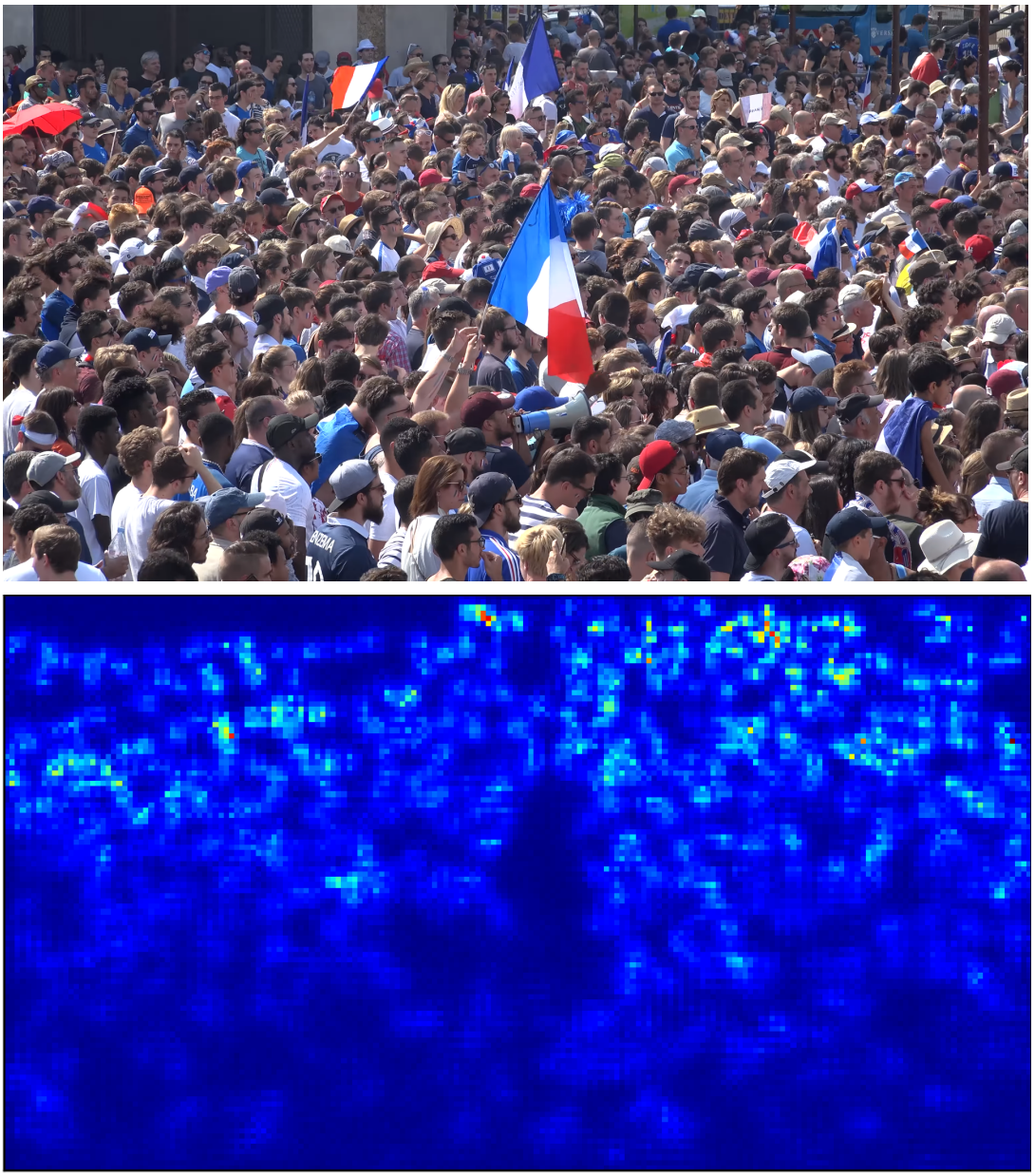


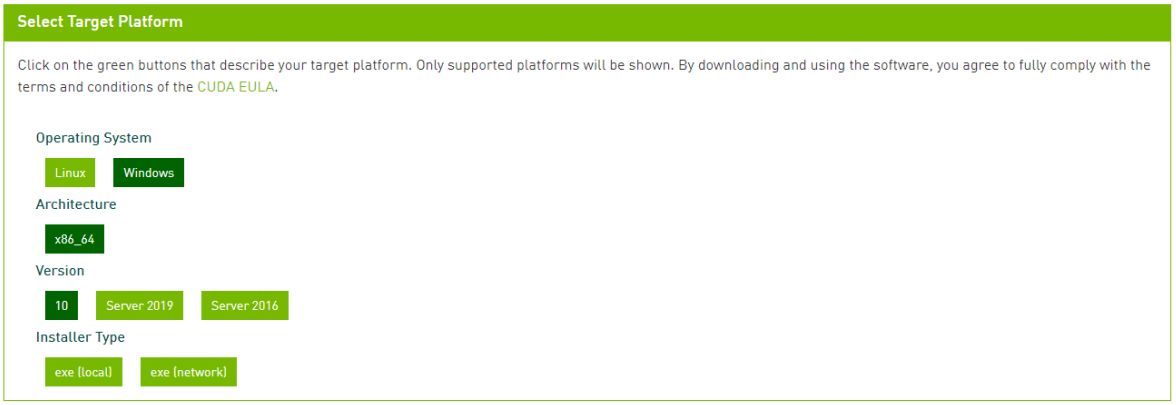
Figure 1: Example of the crowd counting density map

2. Prerequisite

The Crowd Counting application depends on the high computing capacity as the application uses a deep learning neural network to estimate the number of people appearing in an image. The Application needs a **CUDA** enabled **GPU** in order to speed up the compute-intensive algorithms and Cuda Toolkit should be installed prior to the installation of the Crows Counting application.

Setup Cuda Toolkit on Windows 10(Nvidia GPU)

- Download the Cuda toolkit from the link.
- Select the appropriate platform as shown in the below figure.



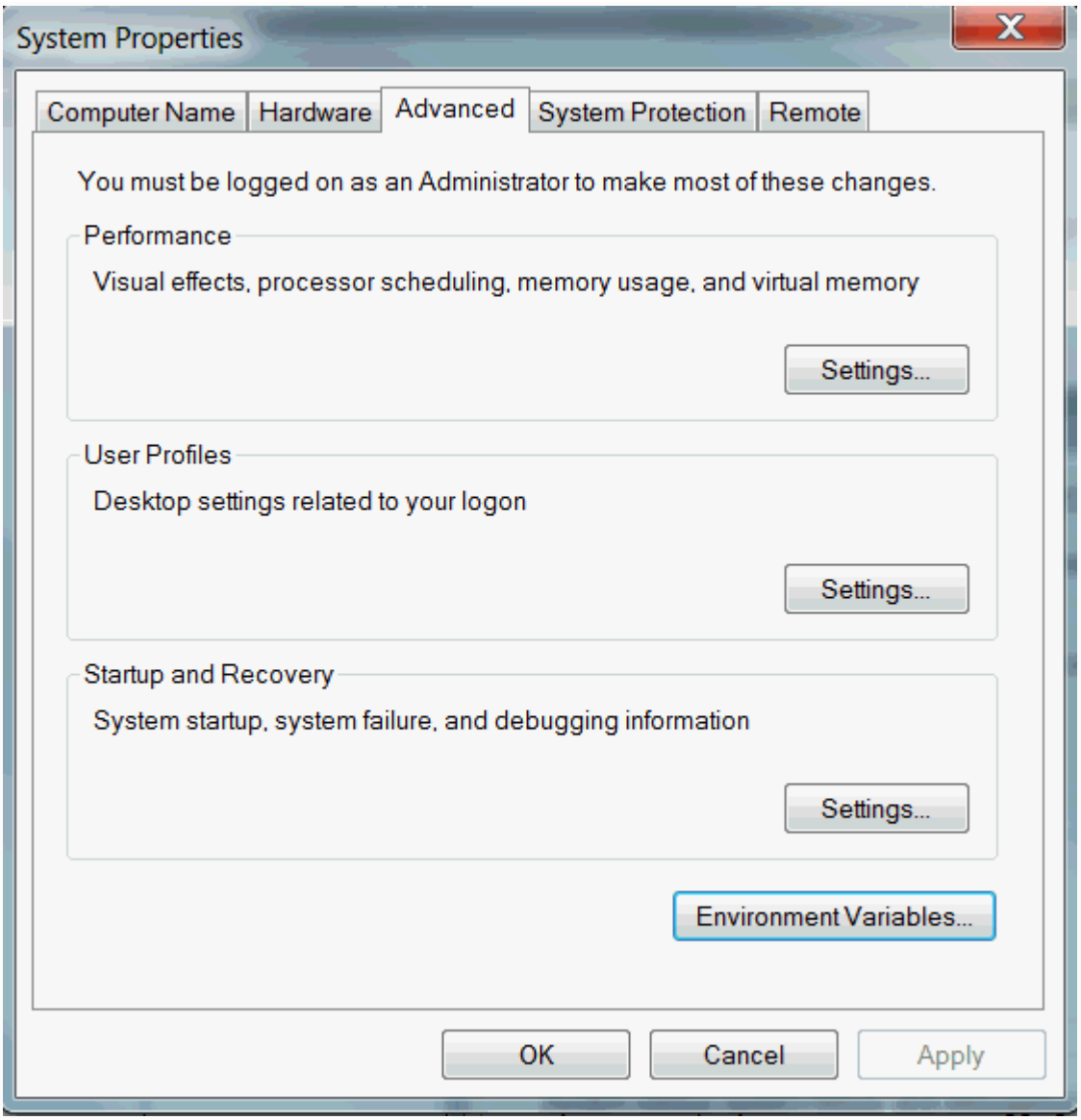
- Run the Cuda toolkit installer.

Setup Crowd Counting Application

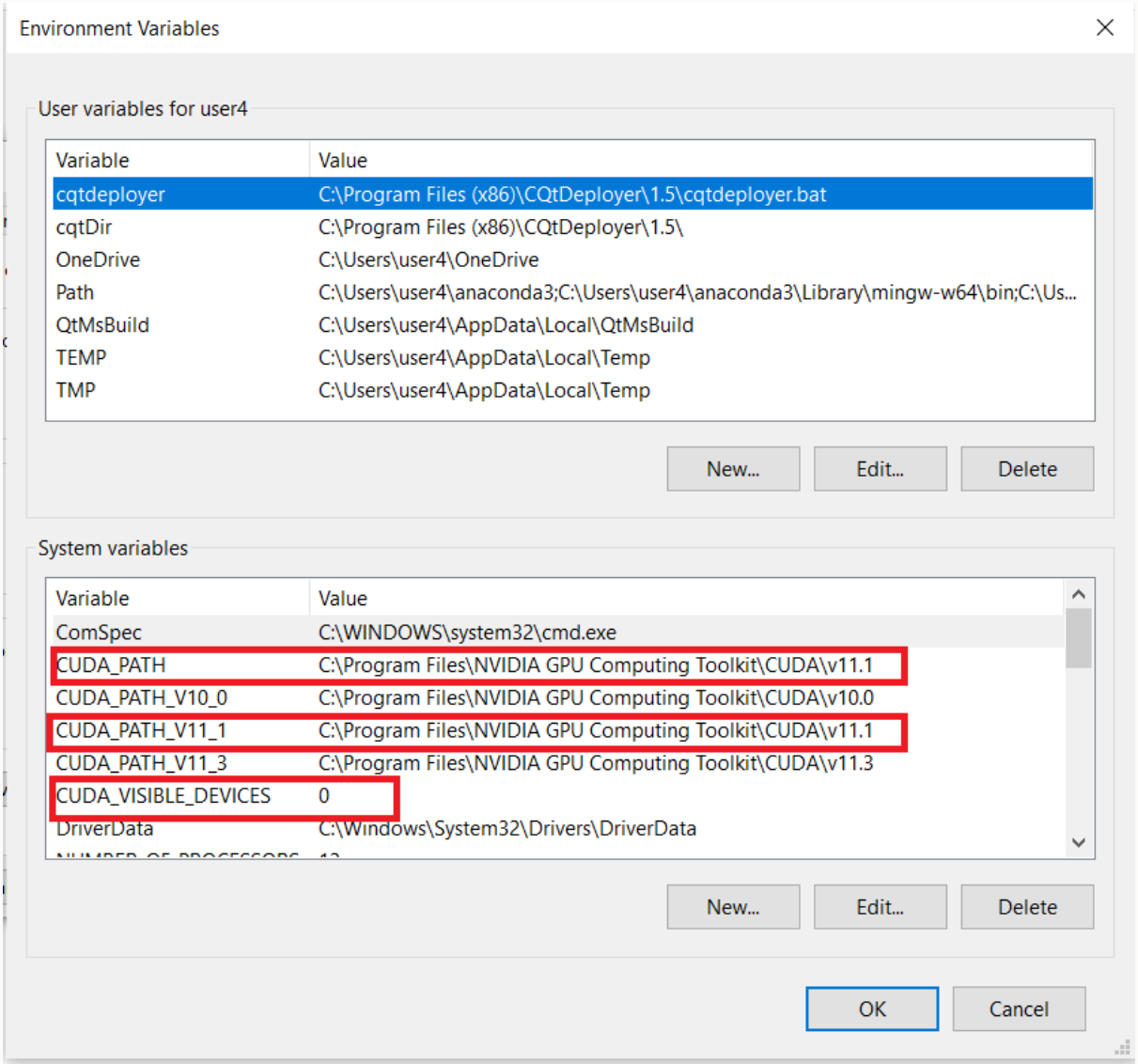
- Run the provided ObjectDetectionInstaller.exe. The Installer will check if the toolkit is already installed or not and abort the installation if no cuda toolkit is installed.
- Installer will prompt for administrator rights during the installation and make sure that proper administrative rights are given for smooth installation.
- The Installer will check dependent modules and install if it is not installed.

How to Verify the Cuda Toolkit Installation

- On the Windows taskbar, right-click the **Windows** icon and select **System**.
- In the **Settings** window, under **Related Settings**, click **Advanced system settings**.
- On the **Advanced** tab, click **Environment Variables**



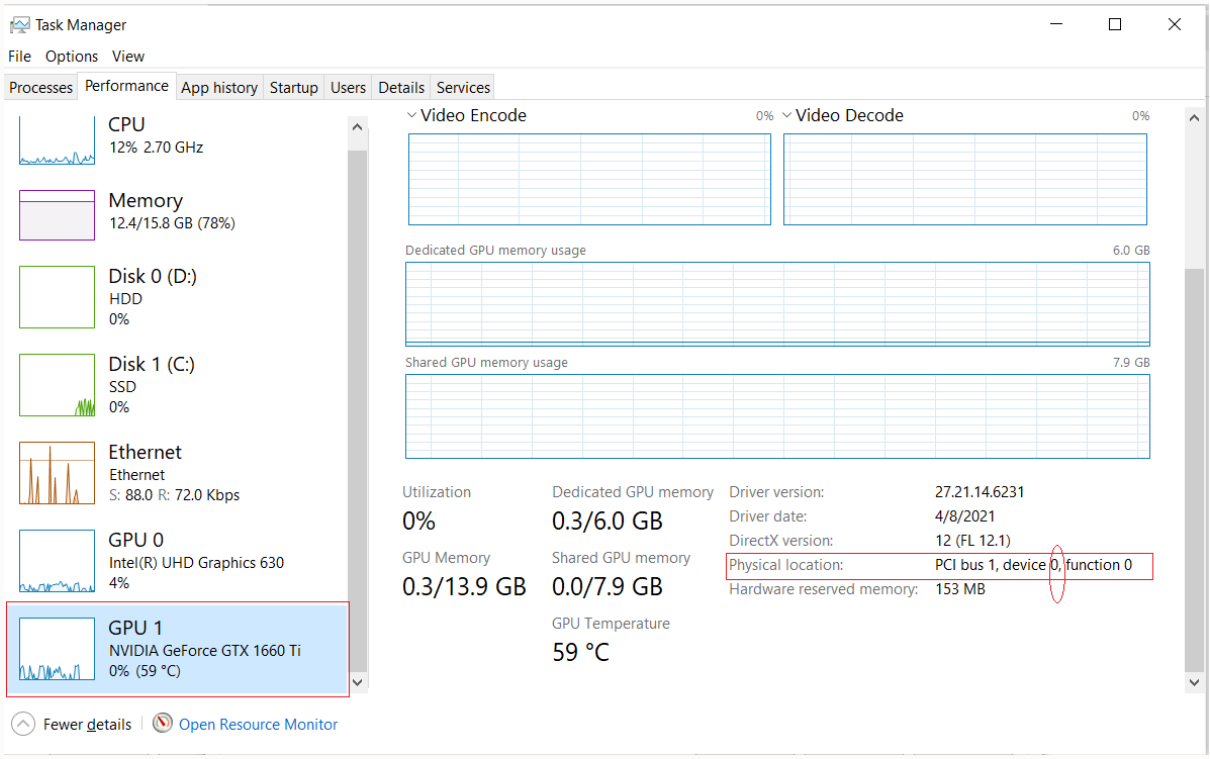
- Ensure that Following Environment variables are set up after the installation as per below figure.



CUDA_PATH : Ensure that it points to the cuda toolkit 11.1 path. Multiple cuda path with specific version will be available If system installed other cuda toolkit versions(CUDA_PATH_V10_0, CUDA_PATH_V11_3 etc); So ensure that your **CUDA_PATH** points to CUDA_PATH_V11_1

CUDA_VISIBLE_DEVICES : The value should be 0 you if you have only one Nvidia GPU

- Open Windows **Task Manager** and click on the **Performance** tab
- Click on the Correct Nvidia GPU and check the Physical Location
  set the CUDA_VISIBLE_DEVICES number based on the device number mentioned in the **Physical location** area, For example below image it is device 0 and CUDA_VISIBLE_DEVICES should be 0.

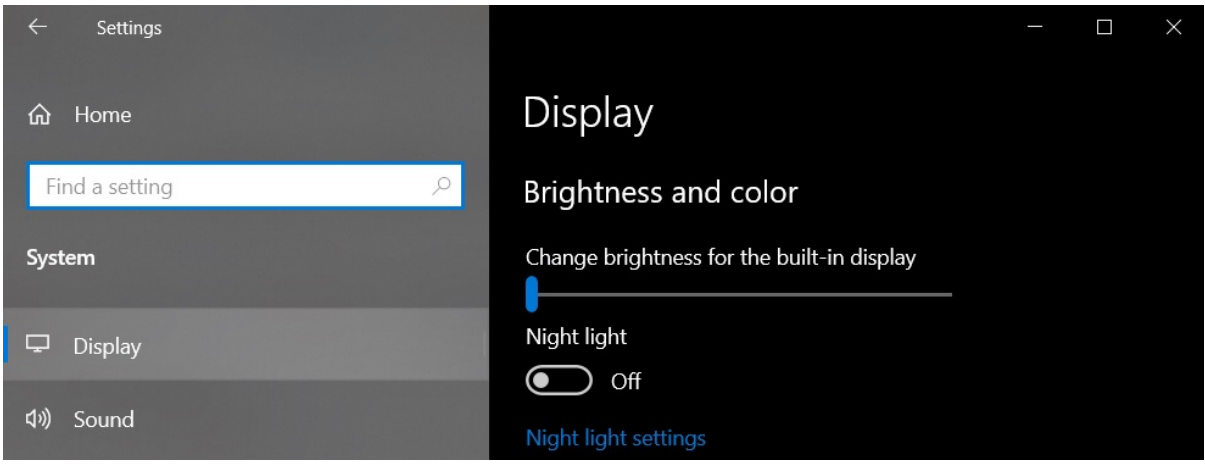


Configure System for Best GPU Performance

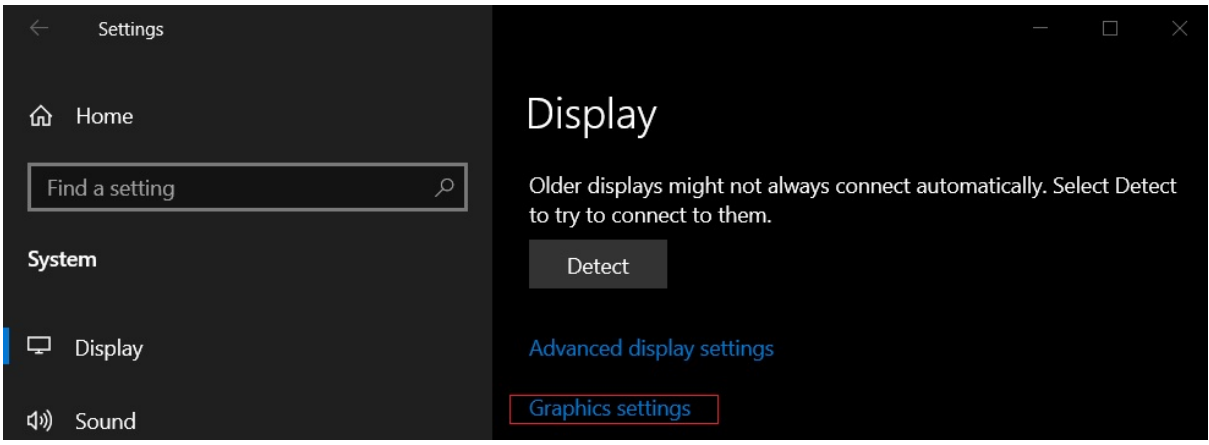
Application performance depends on the System GPU acceleration.Thus ensure that GPU acceleration is enabled and there are no restrictions based on the power mode.

To enable hardware accelerated GPU scheduling on Windows 10, use these steps:

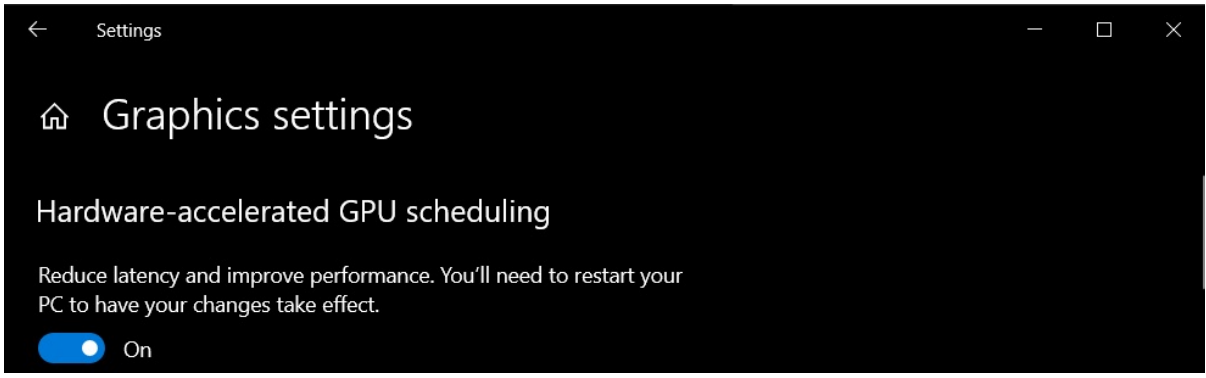
- Open Start Menu and tap on Settings icon.
- In Settings, click on ‘System’ and open the ‘Display’ tab.



- Under the “Multiple Displays” section, select “Graphics settings”.

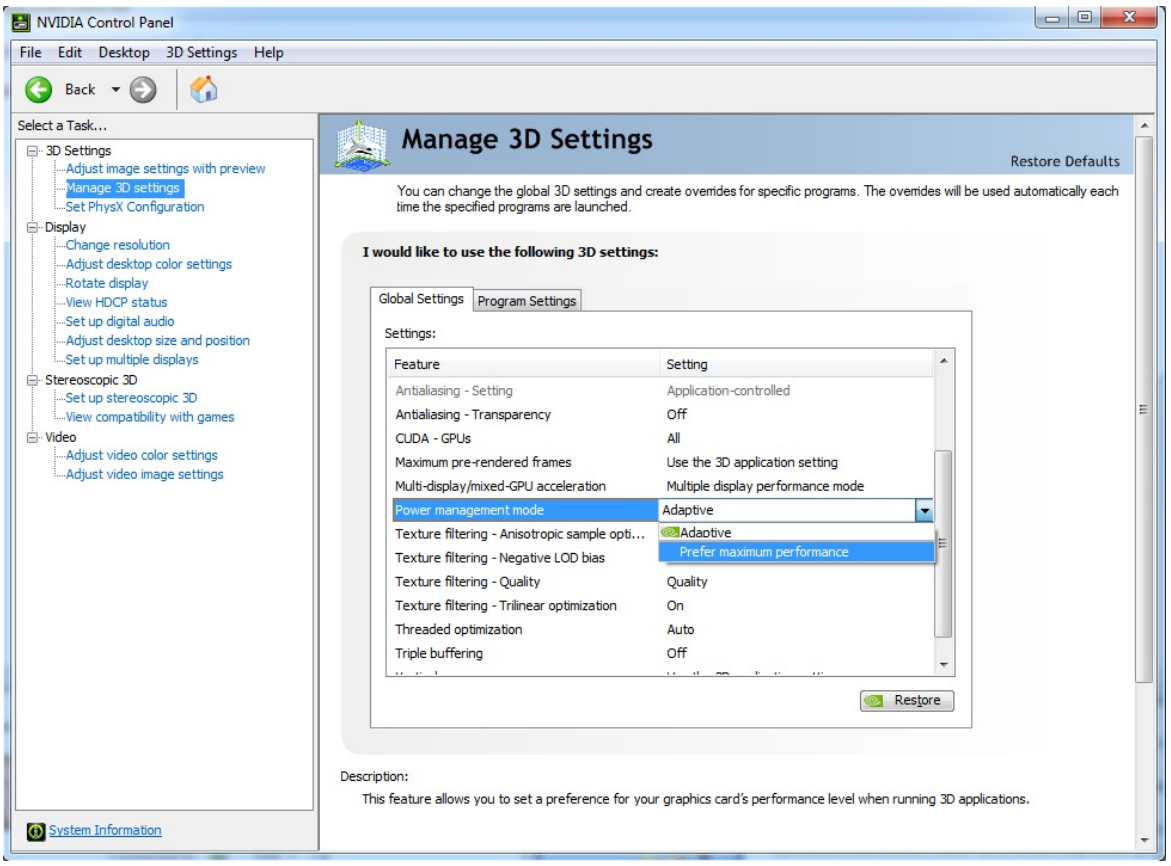


- Turn on “Hardware-accelerated GPU scheduling” option and Restart the system.



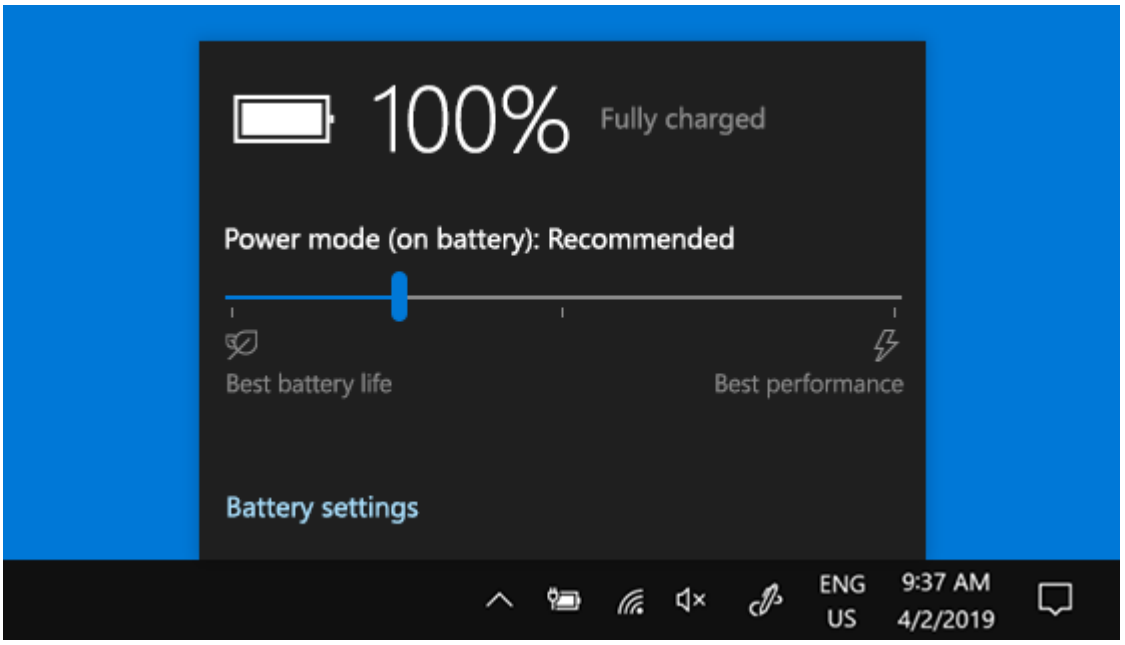
Setting "Power management mode" from Adaptive to Maximum Performance

Setting Power management mode from "Adaptive" to "Maximum Performance" can improve performance in certain applications when the GPU is throttling the clock speeds incorrectly. To change this setting, with your mouse, right-click over the Windows desktop and select "NVIDIA Control Panel" -> from the NVIDIA Control Panel, select the "Manage 3D settings" from the left column -> click on the Power management mode drop down box and select "Prefer Maximum Performance". Click over the "Apply" button at the bottom of the panel to apply the changes.



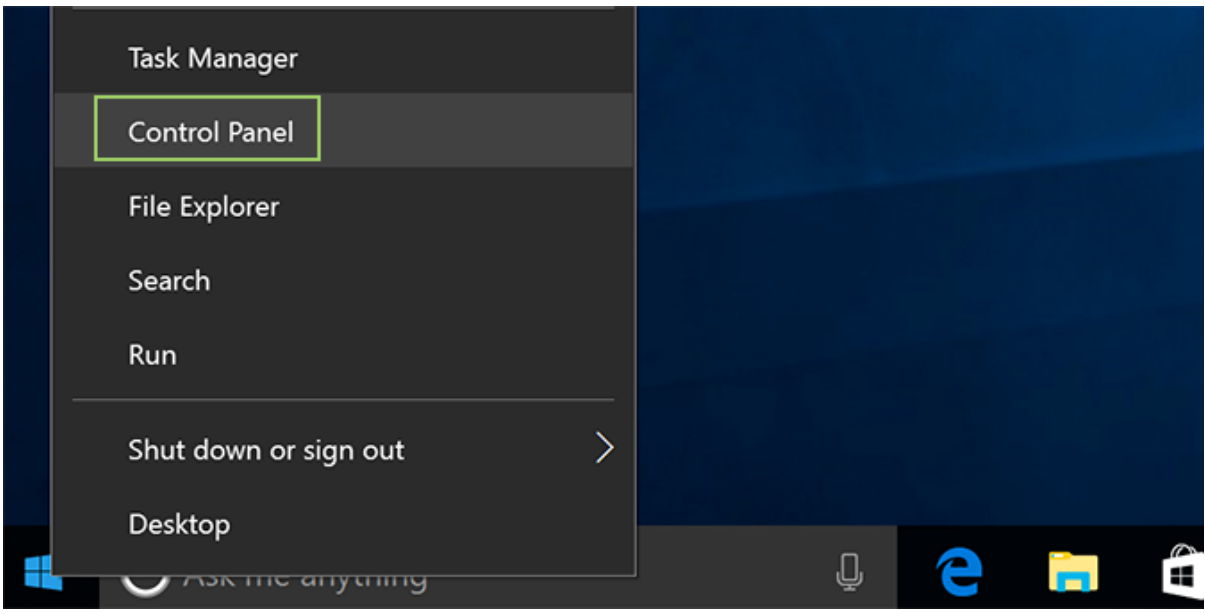
Change the power mode to best performance for your Windows PC

To change the power mode quickly, select the Battery icon on the taskbar, and then drag the slider to the power mode “Best Performance”.

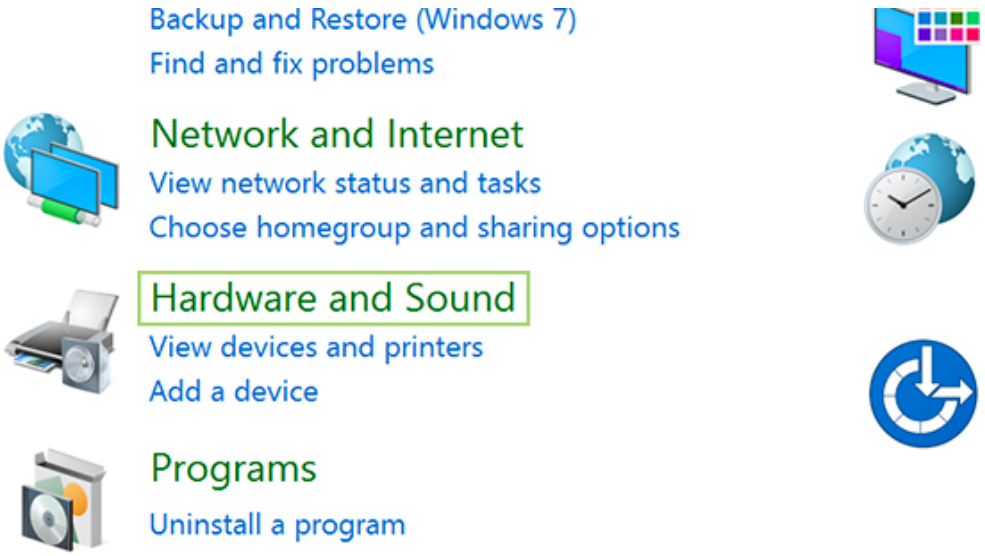


Optional: Settings to Use Maximum CPU Power

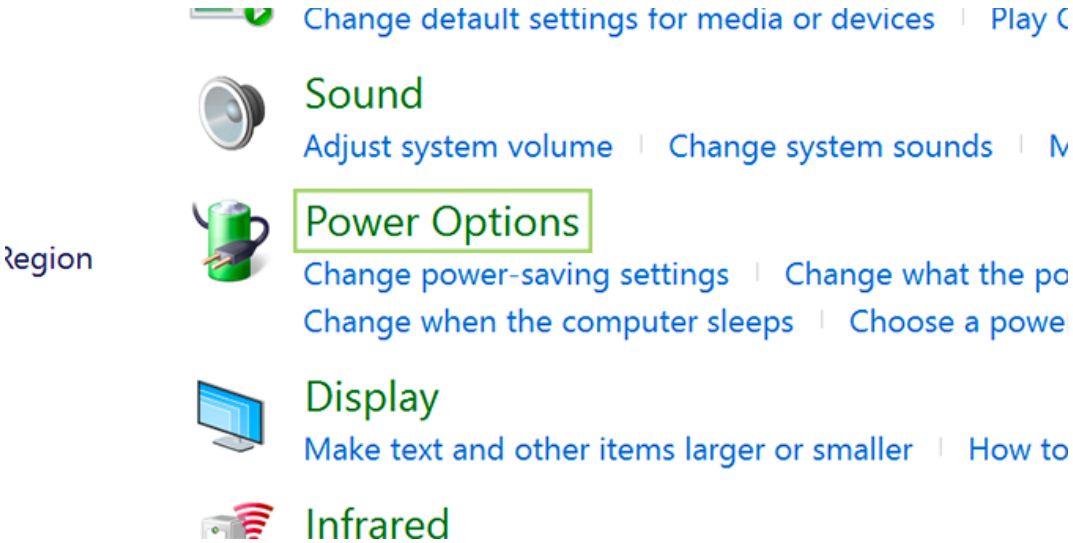
- Right click the Start menu and select the Control Panel.



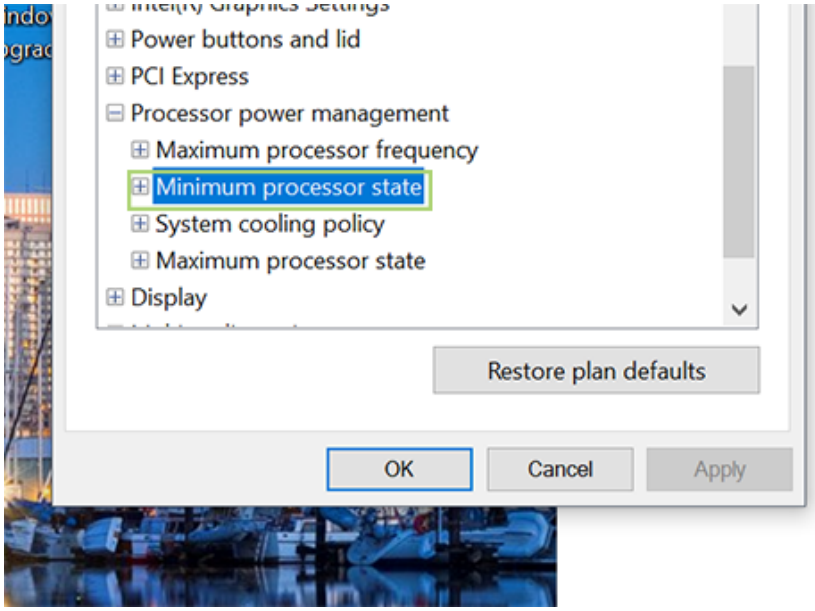
- Click Hardware and Sound.



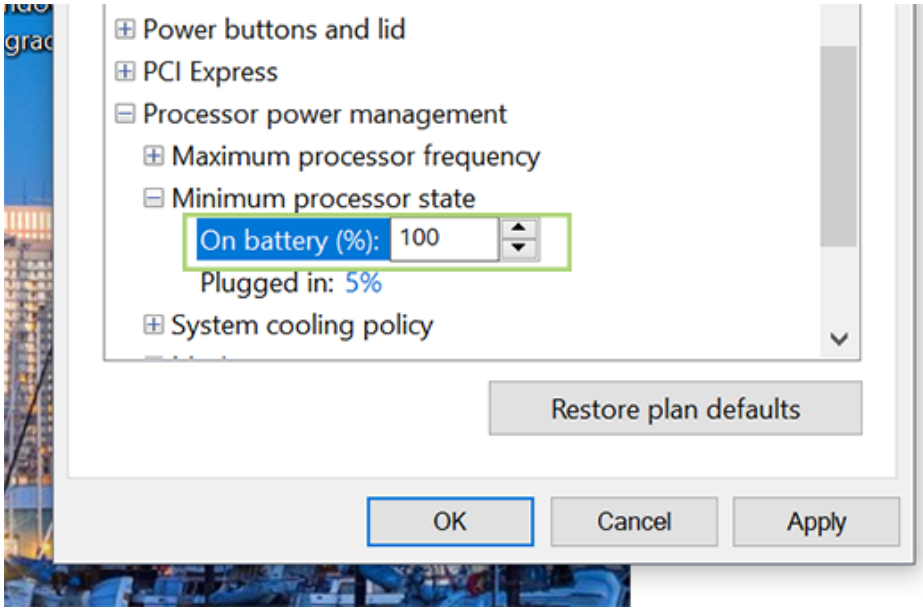
- Select Power Options.



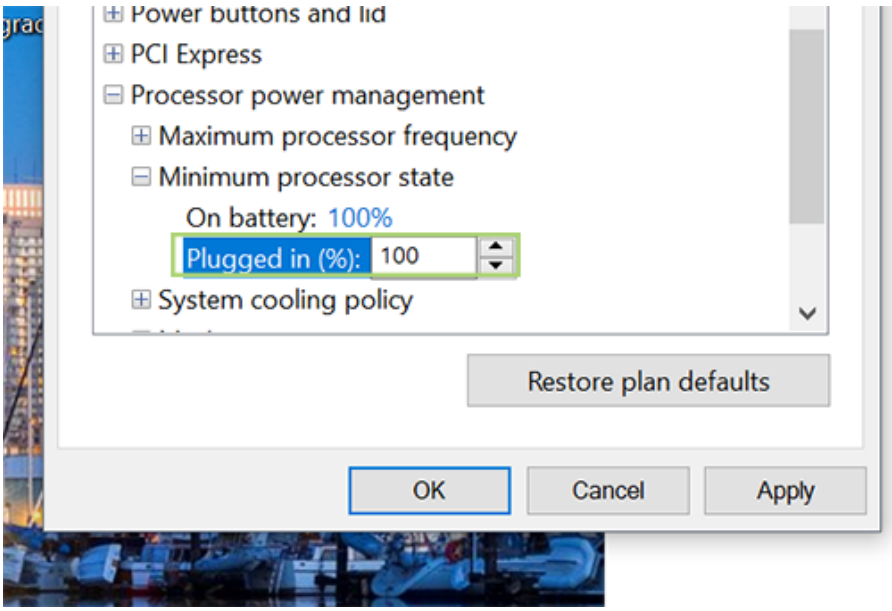
- Find Processor power management and open the menu for Minimum processor state.



- Change the setting for the battery to 100%.



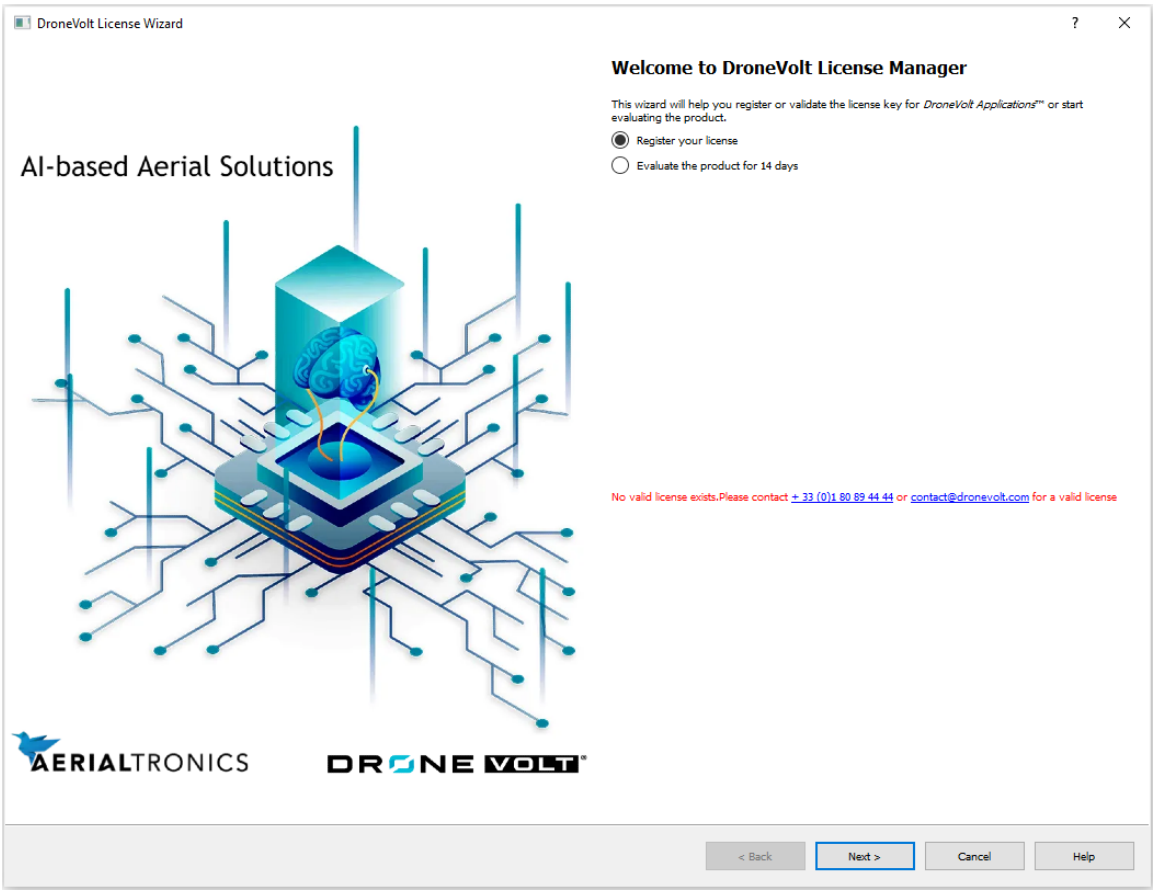
- Change the setting for plugged in to 100%.



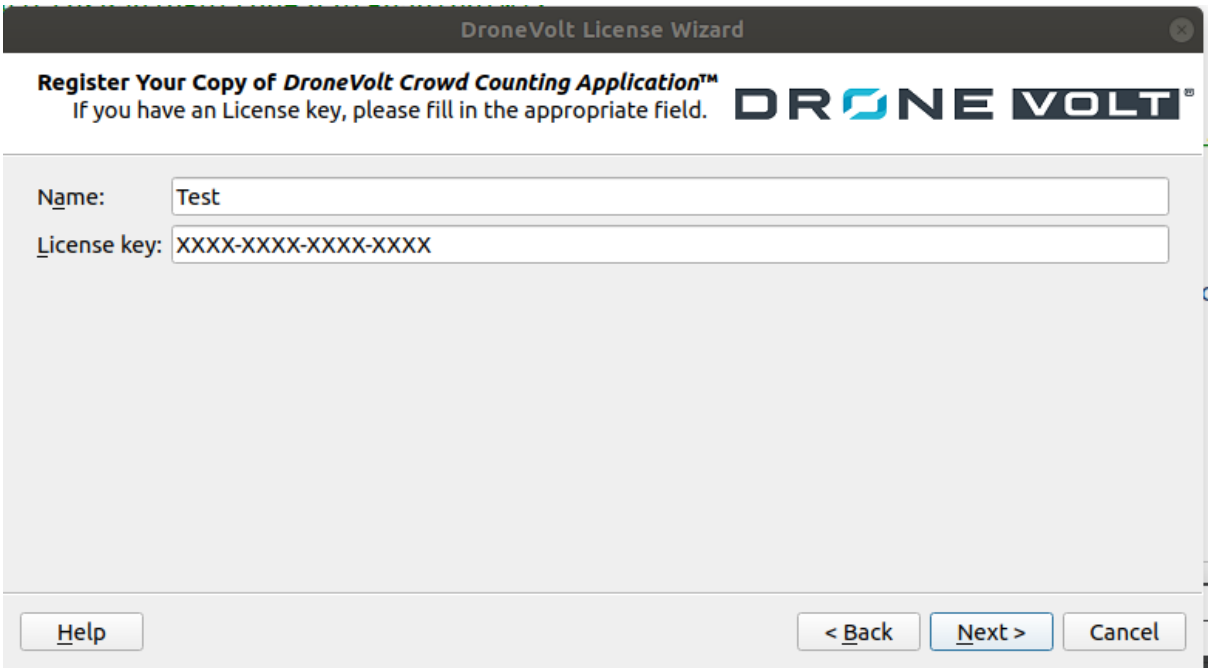
3. Activate License key

A. Start the Crowd Counting Application
Shortcuts will be available either in desktop or Application Menu.

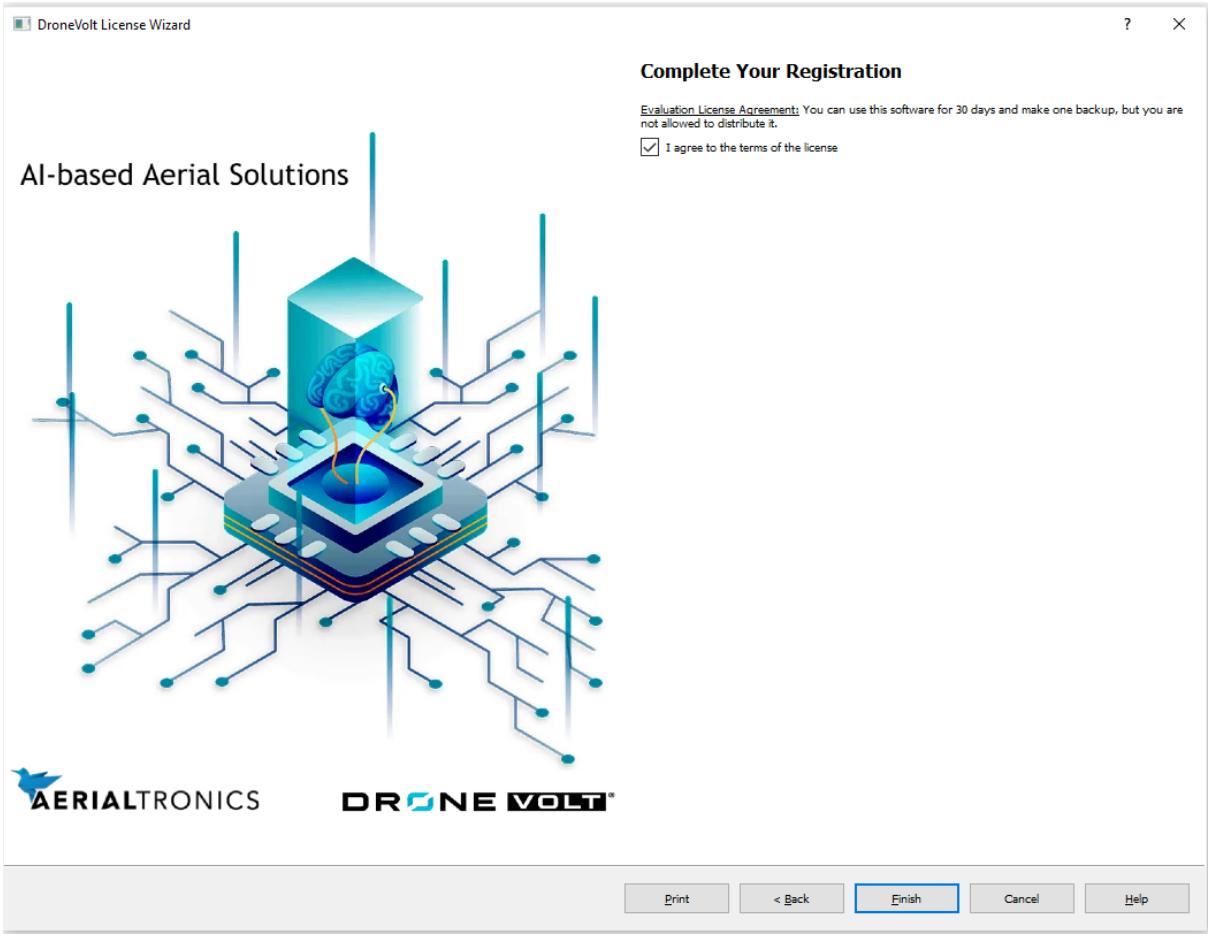
B. Select the “Register your license” option as shown in the diagram below and click “Next”.



C. Enter a valid License Key and Click “Next”.



D. Check the Agreement box and Click on “Finish”.



4. Application Guidelines

A. Start the Crowd Counting Application

The application has launched successfully as shown in the image below.

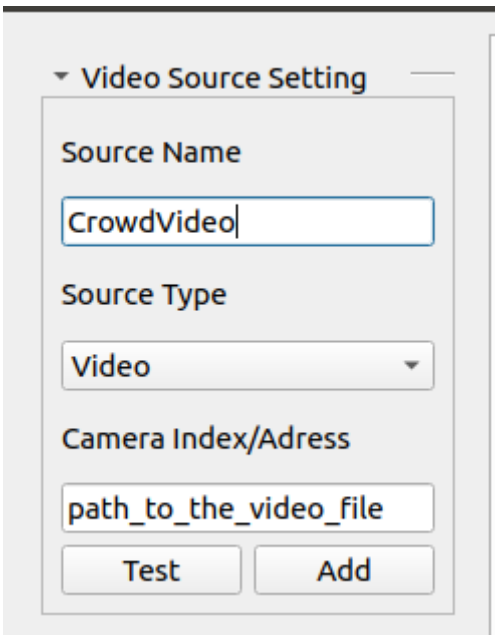


B. Configure the Video Source to the application

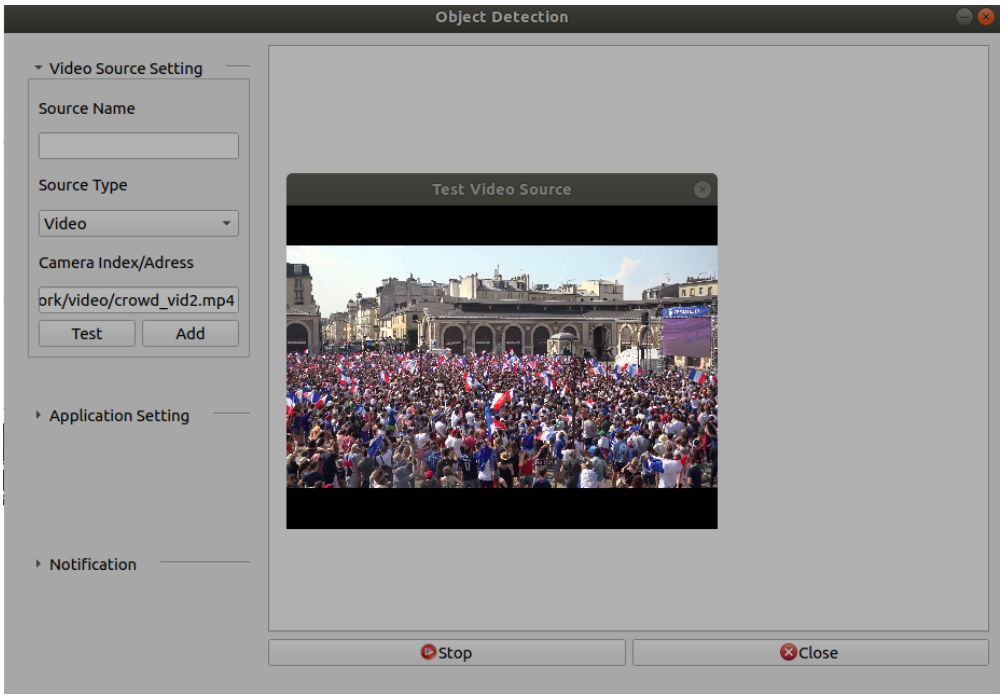
i. Expand the “Video Source Setting” from the left pane of the application user interface.

ii. Configure the Video source with the following details

1. “Source Name” : A unique name for identifying the video source.
2. “Source Type” : Currently 3 sources types are available
    a. “USB”
    b. “RTSP”
    c. “Video”
3. “Camera Index/Address”:
   Camera address will be change based on the source type for for example
   “USB” : “/dev/video0”
   “RTSP”:
   rtsp://user_name:password@IP_address:port_number
   “Video” : /home/user/work/video/crowd_video.mp4



4. Click on the “Test” Button to test the video source.
   A new Window will popup with video streaming as shown below. An image will be displayed if the configuration is valid, otherwise a proper error message will appear.



5. Click on the “Add” button in order to save the configuration in the application. Only Saved video sources can be chosen at the time of running the crowd counting application.

C. “Run” Crowd Counting

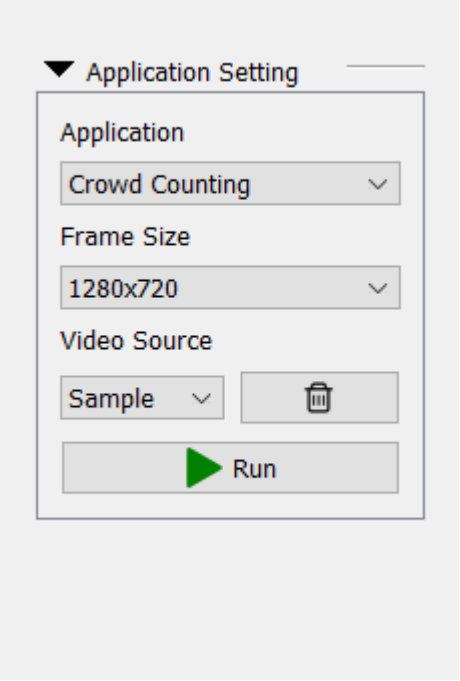
   i. Expand the “Application setting” from the left pane of the application Graphical user interface.



ii. Choose the detection type which user wants to run from the list
eg: “Crowd Counting”.

iii. Select the proper frame size
eg: 1280x720.

iv. Choose the Video source from the list
eg: CrowdVideo , configured in the step 3.B.ii.C



v. Click on the “Run” button to start the crowd counting.
The application will start to process the video frame and processed frames will be displayed on the application window as shown in below image. Besides, Crowd count will be displayed under the Notification section of the application.



vi. Users can click on the “Pause” button from the application user interface in order to pause the video. A “Resume” button will replace the pause button for a paused video so that users can resume the crowd counting process.

vii. Crowd counting will automatically stop once it has finished processing all the frames in the video. Use “Run” button to restart the counting

viii. Click on the “Close” button to close the Video source Tab or stop the video.